SUDONG LEE

Google Scholar ◇ LinkedIn
ORCID: 0000-0002-8928-6070

EDUCATION

Ph.D. student in Robotics, Control, and Intelligent Systems (EDRS) 04.2023 - present
EPFL (École Polytechnique Fédérale de Lausanne) *Lausanne, Switzerland.*
Advisor: Prof. Josie Hughes

M.S. in Mechanical Engineering 03.2019 - 08.2021
Seoul National University *Seoul, Korea.*
Advisor: Prof. Yong-Lae Park
Thesis: Modularized Robotic Skin Sensorized by Fiber Optic Force Sensing for Remote and Autonomous Robot Operation

B.S. in Mechanical Engineering 03.2013 - 02.2019
Korea University *Seoul, Korea.*

RESEARCH EXPERIENCE

CREATE Lab (Computational design of Robots and Embodied Autonomous TEchnologies Laboratory)
- EPFL *Lausanne, Switzerland.*
Doctoral Assistant 04.2023 - present

Research topics:
· Morphological effects of finger bone length variation on manipulation
· Robotic skin enabled by layered structures and soft sensing mechanisms
· Robotic shoulder joint with compliant mechanism and passive stabilization

Soft Robotics Research Center (SRRC)
- Seoul National University *Seoul, Korea.*
Research Associate 09.2022 - 03.2023
Research Assistant 09.2021 - 08.2022

Research topics:
· Soft bending actuator with fiber-jamming variable stiffness and fiber-optic proprioception
· Soft electronics and sensors using selective patterning of conductive Elastomers

Soft Robotics and Bionics Laboratory (SRBL)
- Mechanical Engineering, Seoul National University *Seoul, Korea.*
Graduate Student Researcher 01.2019 - 08.2021

Research topics:
· Multi-funtional soft sensor enabled by heterogeneous sensing mechanisms
· Robotic skin sensorized with fiber bragg gratings (FBGs)
· Environmental adaptability of legged robots with cutaneous inflation and sensation

HONORS AND AWARDS

M.S. Thesis Presentation Award 06.2021
Mechanical Engineering, Seoul National University

Third Place Award for Locomotion Challenge 04.2019
IEEE International Conference on Soft Robotics 2019 (RoboSoft 2019)
Team SRBL (Sudong Lee, G. Shin, J. Kim, M. Choi, Y. Baek, and Y.-L. Park)

Great Honor, Winter 2018 Graduation 02.2019
Korea University

Semester High Honors
Korea University
1st Semester, 2018,
2nd Semester, 2017,
1st Semester, 2015,
2nd Semester, 2014,
1st Semester, 2014,
2nd Semester, 2013,
1st Semester, 2013

SCHOLARSHIPS

Kwanjeong Fellowship — 2nd Semester, 2020, 1st Semester, 2020, 2nd Semester, 2019, 1st Semester, 2019
Kwanjeong Educational Foundation

National Science and Engineering Scholarship — 2nd Semester, 2018, 1st Semester, 2018, 2nd Semester, 2017, 1st Semester, 2015
Korea Student Aid Foundation

Academic Excellence Scholarship — 2nd Semester, 2014
Korea University

Best Honor Scholarship — 1st Semester, 2014
Korea University

PUBLICATIONS

Journal Papers

1. **Sudong Lee**, S.M.H. Sadati, A. Georgopoulou, H. Hauser, F. Vollrath, and J. Hughes, "Spider-Web Inspired 3D Printed Soft Sensor Designs for Physical Information Structuring and Damage Resilience." (*Under review*)
2. **Sudong Lee** and J. Hughes, "A multimodal, multicontact, and scalable soft robot skin enabled by layered structures," *Soft Robotics*, 2026. *(DOI: 10.1177/21695172261485887)*
3. **Sudong Lee**, K. Gilday, and J. Hughes, "Tendon-driven robotic shoulder joint with passive stabilization for large workspace," *IEEE/ASME Transactions on Mechatronics*, 2026. *(DOI: 10.1109/TMECH.2026.3674717)*
4. A. Georgopoulou, M. A. Calvo, L. Lucherini, **Sudong Lee**, J. Hughes, and E. Amstad, "Programmable somatosensory soft robots," *npj Flexible Electronics*, vol. 10, 58, 2026. *(DOI: 10.1038/s41528-026-00558-0)*
5. A. Georgopoulou, **Sudong Lee**, B. Dai, F. Bono, J. Hughes, and E. Amstad, "3D printing of self-healing longevous multi-sensory e-skin," *Communications Materials*, vol. 6, 121, 2025. *(DOI: 10.1038/s43246-025-00839-7)*
6. **Sudong Lee***, J. I. Kim*, Y. Baek, D. Chang, J. Lee, Y. S. Park, D. Lee, and Y.-L. Park, "Fiber-optic force sensing of modular robotic skin for remote and autonomous robot control," *IEEE Transactions on Robotics*, vol. 40, pp. 2373-2389, 2024. *(DOI: 10.1109/TRO.2024.3378178)*
   : Sudong Lee and J. I. Kim are co-first authors.
7. D. Kim, **Sudong Lee**, T. H. Hong, and Y.-L. Park, "Exploration-based model learning with self-attention for risk-sensitive robot control," *npj Robotics*, vol. 1, 7, 2023. *(DOI: 10.1038/s44182-023-00006-5)*
8. J. Kang*, **Sudong Lee***, and Y.-L. Park, "Soft bending actuator with fiber-jamming variable stiffness and fiber-optic proprioception," *IEEE Robotics and Automation Letters*, vol. 8, no. 11, pp. 7344-7351, 2023. *(DOI: 10.1109/LRA.2023.3316075)*
   : Sudong Lee and J. Kang are co-first authors.
9. T. Kim*, **Sudong Lee***, S. Chang, S. Hwang, and Y.-L. Park, "Environmental adaptability of legged robots with cutaneous inflation and sensation," *Advanced Intelligent Systems*, vol. 5, no. 11, 2300172, 2023. *(DOI: 10.1002/aisy.202300172)*
   : Sudong Lee and T. Kim are co-first authors.
   Cover Article: 10.1002/aisy.202370050, Editor's Choice: [Link]
10. Y. Lee, S. Lim, W. J. Song, **Sudong Lee**, S. J. Yoon, J.-M. Park, M.-G. Lee, Y.-L. Park, and J.-Y. Sun, "Triboresistive touch sensing: grid-free touch-point recognition based on monolayered ionic power generators," *Advanced Materials*, vol. 34, no. 19, 2108586, 2022.
    (DOI: 10.1002/adma.202108586)
11. G. Shin*, **Sudong Lee***, and Y.-L. Park, "Selective patterning of conductive elastomers embedded with silver powders and carbon nanotubes for stretchable electronics," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 4983-4990, 2022. *(DOI: 10.1109/LRA.2022.3153707)*
    : Sudong Lee and G. Shin are co-first authors.
12. T. Kim, **Sudong Lee**, T. Hong, G. Shin, T. Kim, and Y.-L. Park, "Heterogeneous sensing in a multifunctional soft sensor for human-robot interfaces," *Science Robotics*, vol. 5, no. 49, eabc6878, 2020. *(DOI: 10.1126/scirobotics.abc6878)*

Conference Papers

1. Q. Fu*, **Sudong Lee***, A. Grillo, J. Arreguit, L. Gevers, J. Hughes, and A. J. Ijspeert, "Polymander II: an amphibious salamander-inspired robot with contact and flow sensors," *2026 IEEE International Conference on Robotics and Automation (ICRA)*, Vienna, Austria, 2026, pp. 5438-5445. *(DOI: 10.1109/ICRA57385.2026.11696137)*
: Sudong Lee and Q. Fu are co-first authors.
2. **Sudong Lee** and J. Hughes, "Morphological and material programability of a hall-effect based soft tactile sensors," *2024 IEEE 7th International Conference on Soft Robotics (RoboSoft)*, San Diego, CA, USA, 2024, pp. 325-331. *(DOI: 10.1109/RoboSoft60065.2024.10521990)*

CONFERENCE PRESENTATIONS

1. A. Georgopoulou, **Sudong Lee**, B. Dai, J. Hughes, and E. Amstad, “Multimodal selective sensory receptors for robotic e-skin,” *2025 IEEE 8th International Conference on Soft Robotics (RoboSoft)*, Lausanne, Switzerland, 2025.
2. **Sudong Lee***, J. I. Kim*, Y. Baek, D. Chang, J. Lee, Y. S. Park, D. Lee, and Y.-L. Park, “Fiber-optic force sensing of modular robotic skin for remote and autonomous robot control,” *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Abu Dhabi, UAE, 2024.
: Sudong Lee and J. I. Kim are co-first authors.
3. J. Kang*, **Sudong Lee***, and Y.-L. Park, "Soft bending actuator with fiber-jamming variable stiffness and fiber-optic proprioception," *2024 IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan, 2024.
: Sudong Lee and J. Kang are co-first authors.
4. G. Shin*, **Sudong Lee***, and Y.-L. Park, "Selective patterning of conductive elastomers embedded with silver powders and carbon nanotubes for stretchable electronics," *2022 IEEE 5th International Conference on Soft Robotics (RoboSoft)*, Edinburgh, United Kingdom, 2022.
: Sudong Lee and G. Shin are co-first authors.

PATENTS

1. T. Kim, **Sudong Lee**, and Y.-L. Park, "Soft Sensor with Multi-Sensing Function," 2022.
(Korea Patent: 10-2384623)

TEACHING EXPERIENCE

Assistive Technology Challenge (ATC) Spring semester, 2025-26
- EPFL
Project coach
Supervisors: Prof. Auke J. Ijspeert, Prof. Silvestro Micera, Prof. Josie Hughes

ME-320: Product development and engineering design Autumn semester, 2025-26
- Mechanical Engineering, EPFL
Teaching assistant, Instructor: Prof. Josie Hughes.

Assistive Technology Challenge (ATC) Spring semester, 2024-25
- EPFL
Project coach
Supervisors: Prof. Auke J. Ijspeert, Prof. Silvestro Micera, Prof. Josie Hughes

ME-320: Product development and engineering design Autumn semester, 2024-25
- Mechanical Engineering, EPFL
Teaching assistant, Instructor: Prof. Josie Hughes.

Interdisciplinary robot competition Spring semester, 2023-24
- EPFL
Team coach, Instructor: Prof. Auke J. Ijspeert.

ME-320: Product development and engineering design Autumn semester, 2023-24
- Mechanical Engineering, EPFL
Teaching assistant, Instructor: Prof. Josie Hughes.

M2794.001700_001: Mechanical product design 1st semester, 2019
- Mechanical Engineering, Seoul National University
Teaching assistant, Instructor: Prof. Yong-Lae Park.

ACADEMIC SERVICE

Reviewer - Journal Papers

IEEE Robotics and Automation Letters (RA-L)
IEEE Transactions on Industrial Electronics (TIE)
IEEE Transactions on Robotics (T-RO)
IEEE/ASME Transactions on Mechatronics (TMECH)
Nature Communications
npj Flexible Electronics
npj Robotics
Soft Robotics

Reviewer - Conference Papers

IEEE International Conference on Robotics and Automation (ICRA)
IEEE International Conference on Soft Robotics (RoboSoft)
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

TECHNICAL STRENGTHS (SKILLS)

Programming Languages	C++, Python, Matlab
Embedded System	Arduino, AVR ATmega, Single-Board Computer (SBC)
Software for System and Robots	ROS, Pybullet
Machine Learning	Pytorch, TensorFlow
Design and Simulation	3D Computer-Aided Design (CAD), Finite Element Analysis (FEA) Software
Fabrication	3D Printing (Additive Manufacturing), Silicone Fabrication

OTHER EXPERIENCE

HackaHealth Hackathon 2026, Lausanne 09.10.2026 - 11.10.2026
Staff & Mentor (project coach).

KSEAS (Korean Scientists and Engineers Association in Switzerland) 02.2025 - present
Operating Committee Member

KORAL (Korean Association in Lausanne) 2023 - present
President (09.2025 - 09.2026)

Republic of Korea Air Force (ROKAF, Military Service) 08.2015 - 08.2017
Staff Sergeant, Honorable discharge.